

4.2 # 1, 2, 3, 5, 17, 18, 21, 22, 23, 24, 39, 40

$$1) \vec{w} = \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix}, A = \begin{bmatrix} 3 & -5 & -3 \\ 6 & -2 & 0 \\ -8 & 4 & 1 \end{bmatrix}$$

If $\vec{w} \in \text{Nul}(A)$, then $A\vec{w} = \vec{0}$, so all you have to do is compute:

$$\begin{bmatrix} 3 & -5 & -3 \\ 6 & -2 & 0 \\ -8 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 3 - 5 + 12 \\ 6 - 6 + 0 \\ -8 + 12 - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

So, **yes**, it is in the null space.

2) Same as #1, just compute $A\vec{w}$.

$$\begin{bmatrix} 5 & 21 & 19 \\ 13 & 23 & 2 \\ 8 & 14 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 25 - 63 + 38 \\ 65 - 69 + 4 \\ 40 - 42 + 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Yes, $\vec{w} \in \text{Nul}(A)$.

3) $A = \begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 1 & 4 & -2 \end{bmatrix}$. To find the Null space, we need to consider $A\vec{x} = \vec{0}$.

$$\begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 1 & 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{aligned} x + 3y + 5z &= 0 \\ y + 4z - 2w &= 0 \end{aligned}$$

$$\vec{x} = \begin{bmatrix} -3y - 5z \\ y \\ \frac{1}{2}y + 2z \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 0 \\ \frac{1}{2} \end{bmatrix} y + \begin{bmatrix} -5 \\ 0 \\ 1 \\ 2 \end{bmatrix} z$$

$x = -3y - 5z$
 $2w = y + 4z \rightarrow w = \frac{1}{2}y + 2z$

Thus, $\text{Nul}(A) = \text{span} \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ \frac{1}{2} \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}$